

Biogeochemical CyclePOD Instances in the Web of Life Simulation Engine: Thermodynamic Mass Conservation Across Planetary Reservoirs

Lead Scientific Communications & Academic Outreach Agent

Web of Life Research Group

<https://github.com/pascalranoroarijaona/WebOfLife>

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Abstract

We present the architectural and thermodynamic implementation of Sprint 006 within the **Web of Life (Gaia)** simulation framework. This sprint introduces dedicated biogeochemical CyclePOD instances for Carbon, Water, Nitrogen, and Phosphorus (`src/cycles/`), establishing explicit stock reservoirs and transfer rate calculations. Grounded in the First Law of Thermodynamics, our framework guarantees closed-loop mass conservation ($\sum \Delta S = 0$) across planetary subsystems (atmosphere, hydrosphere, lithosphere, and biosphere) while tracking exergy dissipation and entropy production driven by solar radiative forcing.

1 Introduction and Thermodynamic Foundations

Planetary ecosystems function as complex, dissipative thermodynamic systems operating far from thermodynamic equilibrium. To simulate planetary-scale homeostasis realistically, computational models must reconcile biological adaptation with macroscopic biogeochemical mass balance. Sprint 006 establishes the core cycling infrastructure for the **Web of Life** engine, bridging thermodynamic free energy dissipation with elemental mass conservation.

1.1 Mass Conservation Invariant

Every CyclePOD instance manages a discrete set of stock reservoirs S_i . The time evolution of each reservoir mass is dictated by incoming and outgoing transfer fluxes $F_{i \rightarrow j}$ per time step Δt :

$$\frac{dS_i}{dt} = \sum_j F_{j \rightarrow i} - \sum_k F_{i \rightarrow k} \quad (1)$$

The global system invariant enforces strict mass conservation across the entire cycle:

$$\frac{d}{dt} \sum_i S_i = 0 \quad (\text{except explicit external stellar inputs/outputs}) \quad (2)$$

2 Concrete CyclePOD Architecture

2.1 Carbon Cycle (`src/cycles/carbon.ts`)

The carbon cycle models the exchange of carbon across atmospheric, oceanic, biospheric, and lithospheric reservoirs. Key fluxes include solar-driven photosynthesis ($F_{\text{atm} \rightarrow \text{bio}}$), terrestrial respiration ($F_{\text{bio} \rightarrow \text{atm}}$), Henry's Law ocean-gas exchange ($F_{\text{atm} \rightleftharpoons \text{ocean}}$), and geological weathering/burial ($F_{\text{litho} \rightleftharpoons \text{ocean}}$).

2.2 Water Cycle (`src/cycles/water.ts`)

The hydrological cycle manages planetary fluid distribution across oceans, atmospheric vapor, ice caps, and terrestrial groundwater. Evaporation and melting fluxes scale directly with incoming solar irradiance (Φ_{solar}), while runoff and precipitation close the hydrological loop.

2.3 Nitrogen Cycle (`src/cycles/nitrogen.ts`)

Governs the transformation of inert atmospheric N_2 into bioavailable forms (NH_4^+ and NO_3^-) via biological fixation and microbial nitrification, balancing assimilation by the biosphere against anaerobic denitrification.

2.4 Phosphorus Cycle (`src/cycles/phosphorus.ts`)

Simulates sedimentary phosphorus dynamics through apatite weathering, soil-phosphate plant uptake, aquatic runoff, and marine sedimentation, establishing the primary long-term bottleneck for primary productivity.

3 Orchestration via EarthPOD

The central **EarthPOD** class (`src/earth_pod.ts`) orchestrates all four cycle modules during the global simulation loop:

$$\text{EarthPOD.step}(\Delta t, \Phi_{\text{solar}}) \implies \{\text{Carbon, Water, Nitrogen, Phosphorus}\}.\text{step}(\Delta t, \Phi_{\text{solar}}) \quad (3)$$

4 Verification and Invariant Enforcement

Unit tests (`tests/sprint_006.test.ts`) verify mass conservation across 1,000 simulation steps under variable solar inputs. Any floating-point divergence exceeding tolerance:

$$\left| \sum S_i(t) - \sum S_i(0) \right| < 1.0 \times 10^{-12} \quad (4)$$

triggers a thermodynamic exception, halting execution to preserve physical validity.

5 Conclusion

Sprint 006 provides a robust, thermodynamically sound biogeochemical foundation for the Web of Life engine. Future sprints will couple these cycles directly with emergent metabolic networks and evolutionary POD dynamics.

Repository Reference: <https://github.com/pascalranoroarijaona/WebOfLife>